

MODULE 4 LYMPHATIC SYSTEM

The Lymphatic System.

THE LYMPHATIC SYSTEM

A reference for the lymphatic system video and lab. This chapter covers lymph and the pathway it follows, the lymphatic vessels, lymph nodes, the lymphatic organs, and common disorders of the system.

BY THE END

1. Define lymph and trace the pathway it follows back to the blood.
2. Describe the structure of lymphatic vessels and a lymph node.
3. Sort the lymphatic organs into primary and secondary.
4. Name common disorders of the lymphatic system.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The lymphatic system

drsrennie-stack.github.io/new-build-bio4-solano/lymphatic-concept-videos.html

The Lymphatic System, an Overview

The lymphatic system is a one-way drainage network. It collects fluid that leaks out of the blood capillaries, filters it, and returns it to the bloodstream, while also housing the body's defenses.

WHAT THE SYSTEM IS AND WHAT IT DOES

Structure	What it is
Lymphatic system	A network of vessels, tissues, and organs that drains tissue fluid and defends the body
Interstitial fluid	The fluid in the spaces between cells, filtered out of the blood capillaries
Lymph	The interstitial fluid once it has entered the lymphatic vessels; a clear, watery fluid
Three functions	Draining excess interstitial fluid back to the blood, absorbing dietary fats from the small intestine, and housing the immune defenses
A one-way system	Lymphatic vessels carry lymph in one direction only, toward the heart; the system has no pump of its own

The Pathway of Lymph

Lymph travels a fixed route from the tissue spaces back to the bloodstream, growing into larger and larger vessels along the way. Follow it in order.

1. **Lymphatic capillaries** blind-ended tubes that take up excess interstitial fluid from the tissue spaces
2. **Lymphatic collecting vessels** larger vessels, with valves, that carry the lymph onward
3. **Lymph nodes** the lymph is filtered as it passes through nodes spaced along the vessels
4. **Lymphatic trunks** collecting vessels merge into large trunks, each draining a major region of the body
5. **Lymphatic ducts** the trunks empty into two ducts: the right lymphatic duct and the larger thoracic duct

6. **Subclavian veins** the ducts return the lymph to the bloodstream where the subclavian and internal jugular veins meet

Lymphatic Vessels

Lymphatic vessels resemble veins, but they begin as blind-ended capillaries and they move fluid without a pump.

THE VESSELS, FROM THE SMALLEST TO THE DUCTS

Structure	What it is
Lymphatic capillaries	The smallest lymphatic vessels, blind-ended and more permeable than blood capillaries
Overlapping endothelial cells	The cells of a lymphatic capillary overlap as one-way flap valves, letting fluid in but not back out
Lacteals	Specialized lymphatic capillaries in the lining of the small intestine that absorb dietary fat
Lymphatic valves	One-way valves in the collecting vessels that keep lymph moving in a single direction
How lymph moves	With no pump, lymph is pushed along by the squeeze of nearby skeletal muscles and by breathing movements
Right lymphatic duct	The smaller duct, draining lymph from the upper right quadrant of the body
Thoracic duct	The larger duct, draining the rest of the body; it begins at a sac called the cisterna chyli

Lymph Nodes

Lymph nodes are the filters of the system. Hundreds of them lie along the lymphatic vessels, cleaning the lymph before it returns to the blood.

THE PARTS OF A NODE AND THE VESSELS THAT SERVE IT

Structure	What it is
Lymph node	A small, bean-shaped organ that filters lymph and houses immune cells
Capsule	The dense connective tissue covering of a node
Cortex	The outer region of a node, packed with lymphocytes in rounded clusters
Medulla	The inner region of a node
Afferent vessels	The lymphatic vessels that carry lymph into a node
Efferent vessels	The lymphatic vessels that carry filtered lymph out of a node, leaving at the hilum
Node clusters	Nodes are especially concentrated in the cervical, axillary, and inguinal regions

HOLD ONTO THIS

Afferent vessels arrive, efferent vessels exit. The same pair of words names flow in and flow out everywhere in the body, so learning it once here pays off again in the kidney and the nervous system.

The Lymphatic Organs

The lymphatic organs are sorted into primary organs, where lymphocytes form and mature, and secondary organs, where lymphocytes meet and respond to invaders. Compare them.

THE PRIMARY AND SECONDARY LYMPHATIC ORGANS			
Organ	Group	Location	Role
Red bone marrow	Primary	The spongy bone of the axial skeleton and limb girdles	Where all lymphocytes form, and where B cells mature
Thymus	Primary	The mediastinum, behind the sternum	Where T cells mature
Lymph nodes	Secondary	Clustered along the lymphatic vessels	Filter the lymph
Spleen	Secondary	The upper left abdomen, behind the stomach	Filters the blood
Tonsils	Secondary	A ring around the entrance to the throat	Guard against inhaled and swallowed pathogens
MALT	Secondary	The mucous membranes of the digestive, respiratory, and other tracts	Guards the body's mucosal surfaces

HOLD ONTO THIS

Only two organs are primary, the red bone marrow and the thymus, because those are the only places lymphocytes are made and matured. Everything else on the list is secondary, a place where a finished lymphocyte meets what it was built to fight.

The Spleen, Thymus, and Tonsils

Three of the lymphatic organs are worth a closer look at their structure.

The spleen

THE SPLEEN AND ITS TWO TISSUES	
Structure	What it is
Spleen	The largest lymphatic organ, in the upper left abdomen behind the stomach
White pulp	The lymphocyte-rich tissue of the spleen that carries out its immune functions
Red pulp	The tissue of the spleen that filters the blood and removes worn-out erythrocytes

The thymus and tonsils

THE THYMUS AND THE RING OF TONSILS

Structure	What it is
Thymus	The primary organ where T cells mature; large in childhood and shrinking after puberty
Tonsils	A ring of lymphatic tissue guarding the throat, including the palatine, lingual, and pharyngeal tonsils
Pharyngeal tonsil	The single tonsil on the posterior wall of the nasopharynx, also called the adenoid

Disorders of the Lymphatic System

When drainage fails or the lymphatic tissue is diseased, the problems show up as swelling or as changes in the nodes and organs. Compare the common ones.

COMPARE THE COMMON ONES BY THE STRUCTURE EACH ONE AFFECTS

Disorder	Structure affected	What it is
Edema	The interstitial space	Swelling from excess interstitial fluid when lymphatic drainage cannot keep up
Lymphedema	The lymphatic vessels	Long-term tissue swelling from blocked or missing lymphatic vessels, sometimes after lymph node removal
Lymphadenopathy	The lymph nodes	Enlarged lymph nodes, often a sign the nodes are fighting an infection
Lymphoma	Lymphatic tissue	A cancer arising in lymphatic tissue, including Hodgkin and non-Hodgkin lymphoma
Splenomegaly	The spleen	An enlarged spleen, which can follow an infection or a blood disorder
Tonsillitis	The tonsils	Inflammation of the tonsils, common in childhood

THE SYSTEM A CLINICIAN READS THROUGH THE SKIN

Swollen glands are **lymphadenopathy**: lymph nodes enlarge as they trap and fight pathogens, which is why a clinician feels for tender nodes in the neck, armpit, and groin during an exam.

Because lymph passes through nodes in order, the lymphatic system also maps how some cancers spread. The first node a region drains into is the **sentinel node**, and checking it tells a surgeon whether a cancer has begun to travel. Removing many nodes can block drainage and leave lasting lymphedema in the limb.

The **spleen** sits just under the lower left ribs. Its soft tissue is easily torn, so a blow to that area can rupture the spleen and cause heavy internal bleeding, since the organ filters a large volume of blood.